Limits & Derivatives

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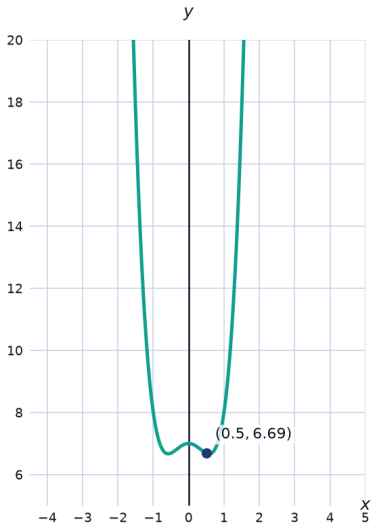
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: One-sided limits**

Using the same piecewise function, does $\lim_{x \to 2} f(x)$ exist? State its value.

2 **[Foundational]** • **Target: The power rule**

Differentiate $f(x) = 3x^4 - 2x^2 + 7$.



3 **[Foundational]** • **Target: The product rule**

Differentiate $f(x) = x^2(3x + 1)$ using the product rule.

4 **[Foundational]** • **Target: Tangent lines**

Find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.

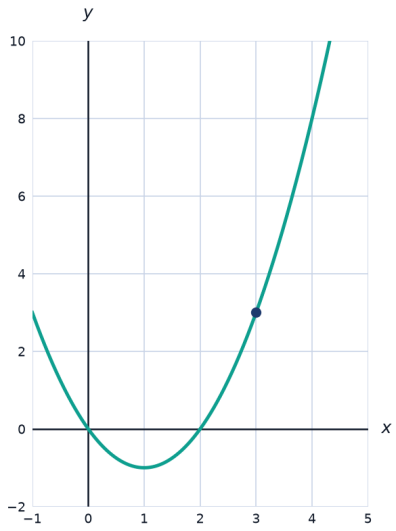
5 **[Foundational]** • **Target: Tangent lines**

Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

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6 [Foundational] • Target: Tangent lines

The graph of $f(x) = x^2 - 2x$ is shown. Find the equation of the tangent line at $x = 3$ (the marked point).



7 [Foundational] • Target: Tangent lines

Find the x-coordinate(s) where the tangent to $f(x) = x^3 - 3x$ is horizontal.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Increasing, decreasing, and the second derivative

For $f(x) = x^2 - 4x + 1$, find $f'(x)$ and determine the interval where f is increasing.

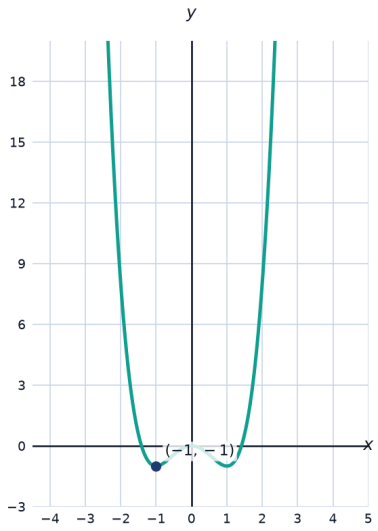
9 [Intermediate] • Target: Increasing, decreasing, and the second derivative

For $f(x) = -x^2 + 6x$, determine the interval where f is decreasing.

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10 [Intermediate] • Target: Increasing, decreasing, and the second derivative

Find the second derivative of $f(x) = x^4 - 2x^2$.



11 [Intermediate] • Target: Word problems: rates of change

This question has two parts. The height of a ball thrown upward is given by $h(t) = -5t^2 + 20t + 2$ (in meters, t in seconds).

a Find the velocity function $h'(t)$.

b Find the velocity at $t = 1$ second, and state whether the ball is rising or falling.

12 [Intermediate] • Target: Word problems: rates of change

This question has two parts. A company's cost to produce x items is $C(x) = 0.01x^2 + 5x + 200$ (in dollars).

a Find the marginal cost function $C'(x)$.

b Find the marginal cost at $x = 100$ items, and interpret its meaning.

13 [Intermediate] • Target: MCQ — Evaluating limits by factoring (the 0/0 form)

Differentiate the function $f(x) = (2x + 1)^4$ using the Chain Rule.

14 [Intermediate] • Target: MCQ — Limits at infinity

Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{6x^2 - 5}{2x^2 + 7x}$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: MCQ — One-sided limits • Cross-Lesson: 2.01 Functions & Transformations

Find the slope of the tangent line to the curve $y = 2x^3 - 5x + 1$ at $x = 2$.

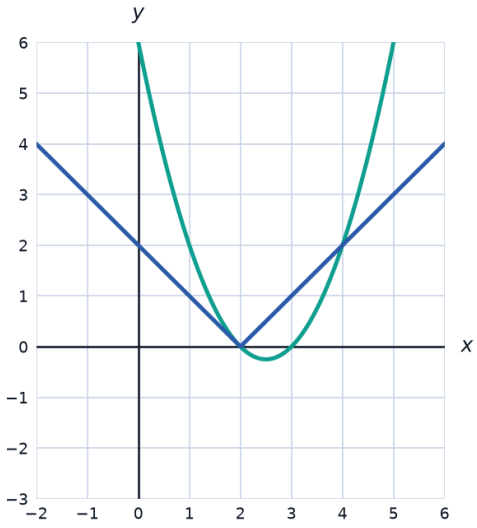
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16 [Challenging] • Target: MCQ — The power rule • Cross-Lesson: 3.01 Sequences & Series

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

17 [Challenging] • Target: MCQ — The product rule • Cross-Lesson: 4.01 Trigonometry

Given the functions $f(x) = x^2 - 5x + 6$ and $g(x) = |x - 2|$, find the value of $f(g(4))$.



18 [Challenging] • Target: MCQ — The chain rule • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

What is the measure of each interior angle in a regular octagon (8-sided polygon)?

19 [Challenging] • Target: MCQ — Derivatives of trigonometric and exponential functions • Cross-Lesson: 7.01 Integration & Applications

Find the slope of the tangent to $y = x^2$ at $x = 2$.

20 [Challenging] • Target: MCQ — Tangent lines • Cross-Lesson: 8.01 Exponentials & Logarithms

Find the area bounded by the curve $y = 3x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 2$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — One-sided limits

Using the same piecewise function, find $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

a. 4; does not exist

b. 4; exists

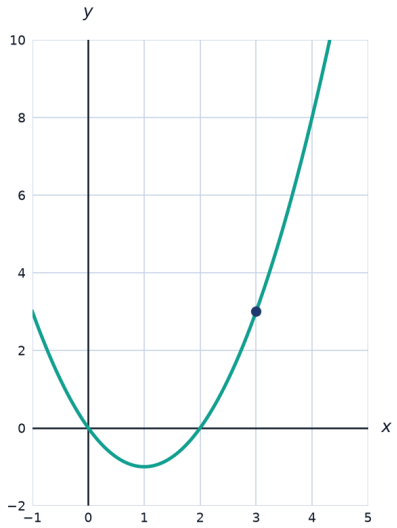
c. 2; exists

d. 2; does not exist

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22 [Foundational] • Target: MCQ — Tangent lines

The graph of $f(x) = x^2 - 2x$ is shown, with the point at $x = 3$ marked. Find the equation of the tangent line there.



a. $y = 3x - 9$

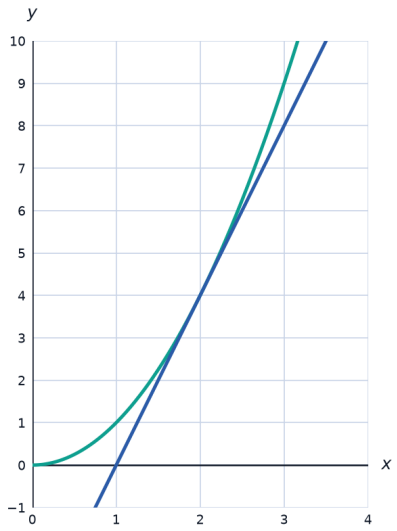
b. $y = 4x - 3$

c. $y = 4x + 9$

d. $y = 4x - 9$

23 [Foundational] • Target: MCQ — Tangent lines

The graph shows $f(x) = x^2$ and its tangent line at $x = 2$. Find the slope of the tangent line by reading the graph.



a. 4

b. 2

c. 1

d. 8

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24 [Foundational] • Target: MCQ — Word problems: rates of change

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the ball's velocity at $t = 2$ seconds.

a. 10

b. -10

c. 30

d. 20

25 [Foundational] • Target: MCQ — Word problems: rates of change

A company's cost to produce x items is $C(x) = 0.02x^2 + 8x + 150$ (in dollars). Find the marginal cost at $x = 100$.

a. 8 dollars

b. 4 dollars

c. 20 dollars

d. 12 dollars

26 [Foundational] • Target: MCQ — Word problems: rates of change

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the time at which the ball reaches its maximum height.

a. $t = 3$

b. $t = 5$

c. $t = 6$

d. $t = 2$

27 [Foundational] • Target: The chain rule

For $f(x) = -x^2 - 3$, determine the interval where f is increasing.

a. $x < 0$

b. $x > 2$

c. $x < 2$

d. $x > 0$

28 [Foundational] • Target: Derivatives of trigonometric and exponential functions

■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■■■ $f(x) = x^2 \sin x$■ Product rule derivative of $x^2 \sin x$?

a. $2x \sin x + x^2 \cos x$

b. $2x \cos x$

c. $x^2 \cos x - 2x \sin x$

d. $2x \sin x$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Tangent lines

■■ ■■■■■ ■■■■■■■ ■■■■■■■■ $f(x) = \frac{x}{x+1}$■ Quotient rule derivative of $\frac{x}{x+1}$?

a. $\frac{1}{(x+1)^2}$

b. $\frac{2x+1}{(x+1)^2}$

c. $\frac{-1}{(x+1)^2}$

d. 1

30 [Intermediate] • Target: Increasing, decreasing, and the second derivative

■■ ■■■■ ■■■■■■■■ $\lim_{x \to 2} (3x^2 - 4x + 1)$■ Evaluate limit by direct substitution.

a. 5

b. 4

c. 7

d. 1

31 [Intermediate] • Target: Word problems: rates of change

■■■ ■■■■ $f(x) = x^3 - 6x^2 + 5$ ■■■■■ ■■■■ $g(x)$ ■■■ ■■■ ■■■■■■■■ ■■■■■■■ ■■■ $x = 4$ ■■■ $f(x)$■ ■■■■■■ $g(4) = f(4)$■ ■■■ ■■ ■■■■ $g'(4)$■ If $f(x) = x^3 - 6x^2 + 5$ and another function $g(x)$ has the same first derivative at $x = 4$ as $f(x)$, and also $g(4) = f(4)$, what is the value of $g'(4)$?

a. 11

b. -19

c. 5

d. -7

32 [Intermediate] • Target: MCQ — Evaluating limits by direct substitution

■■■ ■■■ ■■■■ ■■■■■ ■■■■■■■ $f(x) = x^3 - 3x$ ■■■■■■■■ ■■■ ■■ ■■■ x ■■■■■■■■■ Find x where tangent line is horizontal.

a. ± 1

b. 0 ■■■

c. $\pm\sqrt{3}$

d. 3 ■■■

33 [Intermediate] • Target: MCQ — Evaluating limits by factoring (the 0/0 form)

■■ ■■■■■■■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■■■ $f(x) = x^3 - 3x$ ■■■■■■■■■ Intervals where $x^3 - 3x$ is increasing?

a. $(-\infty, -1) \cup (1, \infty)$

b. $(-1, 1)$

c. $(0, \infty)$

d. $(-\infty, 0)$

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34 **[Intermediate]** • **Target: MCQ — Limits at infinity**

■■ ■■■■■ ■■■■■■ $G(x) = \int_0^{x^2} \sin(t)\,dt$ ■■■■■■■ ■■■ x■ Derivative using Leibniz rule / chain rule?

a. $2x\sin(x^2)$

b. $\sin(x^2)$

c. $\cos(x^2)$

d. $2x\cos(x^2)$

35 **[Intermediate]** • **Target: MCQ — One-sided limits**

■■ ■■■■■ ■■■■■■ ■■■■■■■ $f(x) = 15$■ Derivative of constant function $f(x) = 15$?

a. 0

b. 1

c. 15

d. x

36 **[Intermediate]** • **Target: MCQ — The power rule**

■■ ■■■■■ ■■■■■■ $f(x) = 7x^4$■ Derivative of $f(x) = 7x^4$?

a. $28x^3$

b. $21x^3$

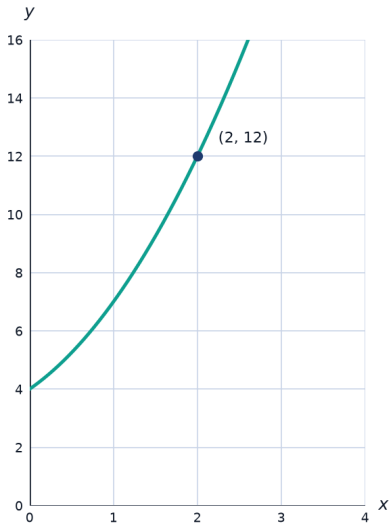
c. $28x^4$

d. $7x^3$

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37 **[Intermediate]** • **Target: MCQ — The product rule**

[redacted] $f(x) = \frac{x^3 - 8}{x - 2}$. [redacted] $f(2)$ [redacted] $f(x)$ [redacted] $x = 2$[redacted] [redacted] Consider the function $f(x) = \frac{x^3 - 8}{x - 2}$. If $f(2)$ is defined such that the function $f(x)$ is continuous at $x = 2$, what is the value of $f(2)$?



a. 12

b. 10

c. 8

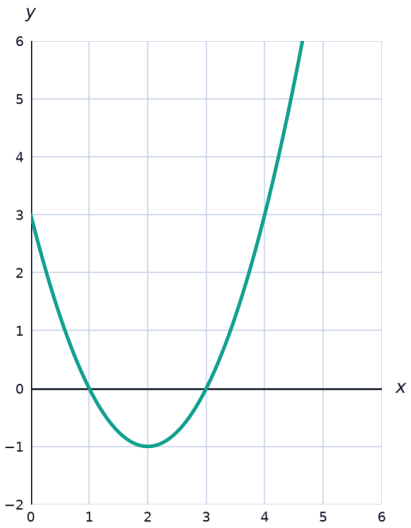
d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — The chain rule • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

[redacted] $f(x) = x^2 - 4x + 3$. [redacted] x [redacted] 3. Let the function be $f(x) = x^2 - 4x + 3$. Find the limit of the function as x approaches 3.



a. 0

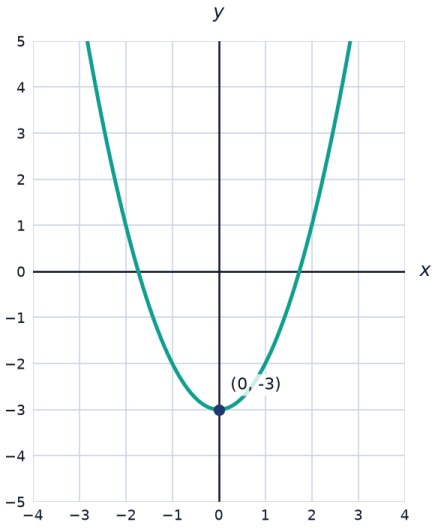
b. 3

c. 6

d. 9

39 [Challenging] • Target: MCQ — Derivatives of trigonometric and exponential functions • Cross-Lesson: 7.01 Integration & Applications

[redacted] $f(x) = x^2 - 3$[redacted] [redacted] x [redacted] If $f(x) = x^2 - 3$, find the critical point and classify it.



a. x=0 (local maximum)

b. x=1 (local minimum)

c. x=-1 (local minimum)

d. x=0 (local minimum)

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40 [Challenging] • Target: MCQ — Tangent lines • Cross-Lesson: 8.01 Exponentials & Logarithms

[illegible] $f(x) = \frac{x^2 + ax + 4}{x - 2}$[illegible] [illegible] a [illegible]. [illegible] [illegible] $\lim_{x \to 2} f(x)$ [illegible] [illegible] [illegible] a[illegible] Given the function $f(x) = \frac{x^2 + ax + 4}{x - 2}$, where a is a constant. If $\lim_{x \to 2} f(x)$ exists, what is the value of a?

a. -4

b. 2

c. 4

d. -2

41 [Challenging] • Target: MCQ — Increasing, decreasing, and the second derivative • Cross-Lesson: 9.01 Statistics & Probability

[illegible] [illegible] [illegible] $f(x) = 3x^2 + 5x - 2$ [illegible] [illegible] x [illegible] 1[illegible] [illegible] [illegible] [illegible] To find the limit of the function $f(x) = 3x^2 + 5x - 2$ as x approaches 1, what is the value?

a. 10

b. 6

c. 7

d. 9

42 [Challenging] • Target: MCQ — Word problems: rates of change • Cross-Lesson: 1.01 Algebra & Equations

[illegible] [illegible] [illegible] $f(x) = \frac{ax^2 - 5x + 3}{x - 1}$ [illegible] [illegible] [illegible] $x = 1$[illegible] [illegible] [illegible] a. If the function $f(x) = \frac{ax^2 - 5x + 3}{x - 1}$ has a limit as $x \to 1$, find the value of a.

a. 1

b. 2

c. 3

d. 5

43 [Challenging] • Target: Evaluating limits by direct substitution • Cross-Lesson: 2.01 Functions & Transformations

[illegible] [illegible] $f(x) = 3x + 5$ [illegible] [illegible] x [illegible] 2 [illegible]:\ The limit of the function $f(x) = 3x + 5$ as x approaches 2 is:

a. 8

b. 9

c. 11

d. 13

44 [Challenging] • Target: Evaluating limits by factoring (the 0/0 form) • Cross-Lesson: 3.01 Sequences & Series

[illegible] [illegible] [illegible] [illegible] [illegible] $f(x) = -x^2 + 4x$: Local max of $f(x) = -x^2 + 4x$.

a. $x = 1$

b. $x = 2$

c. $x = 4$

d. $x = 0$

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45 [Challenging] • Target: Limits at infinity • Cross-Lesson: 4.01 Trigonometry

■■ ■■■■ ■■■■■■■■ $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$■ Evaluate limit involving radical.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. 0

d. 4

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